

Triple Diffie-Hellman Key Exchange (3DH)



Ik_A

$ECDH_1$

$ECDH_2$

Ik_B



Ek_A

$ECDH_3$

SPk_B

$ECDH(Ik_A, SPk_B)$

$ECDH(Ek_A, SPk_B)$

$k_{\text{root}} = \text{HKDF}(ECDH_1 , ECDH_2 , ECDH_2)$

$ECDH(Ek_A, Ik_B)$

Double Ratchet Protocol

- The root key k_{root} **does not encrypt messages**
 - Use **for deriving a series of keys** $k_1, k_2, k_3 \dots$ that will be used to encrypt (send) and decrypt (receive) each message m_1, m_2, m_3
- ➔ **Double Ratchet Key Derivation Protocol**
(more details in bonus challenge)